ForecastLabAI — forecasting model families

11 models · one interface · split by requires_features: target-only vs feature-aware

source: ForecastLabAI @79e8dce · app/features/forecasting/models.py · schemas.py · app/core/config.py

ModelConfig · schemas.py

```
{"model_type": "seasonal_naive",
 "season_length": 7}
```

model_factory(config)
if / elif on model_type

returns

BaseForecaster (ABC)

```
fit(y, X=None)              → self
predict(horizon, X=None)    → np.ndarray
get_params()                → dict
set_params(**params)        → self
```

random_state = 42 → deterministic fits

requires_features: ClassVar[bool]

ForecastingService branches on requires_features — not isinstance(), not model_type

flag gate in model_factory (3 models, settings default False)

```
elif model_type == "lightgbm":
    if not settings.forecast_enable_lightgbm:
        raise ValueError("LightGBM is not enabled. …")
```

forecast_enable_random_forest · forecast_enable_lightgbm · forecast_enable_xgboost

requires_features ?

requires_features = False
target-only · fit(y) · 6 baselines

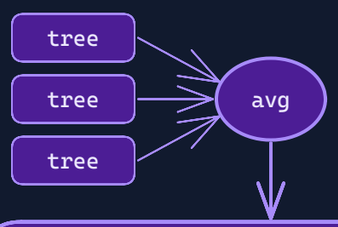
- **naive · NaiveForecaster**
  ŷ[t+h] = y[t] — repeat the last value
- **seasonal_naive · SeasonalNaiveForecaster**
  ŷ[t+h] = y[t+h−m], m = season_length (7)
- **moving_average · MovingAverageForecaster**
  mean of the last window_size (7) values
- **weighted_moving_average · WeightedMovingAverageForecaster**
  linear | exponential weights — recent values weigh more
- **seasonal_average · SeasonalAverageForecaster**
  mean per seasonal position (season_length 7 = weekly)
- **trend_regression_baseline · TrendRegressionBaselineForecaster**
  Ridge on an elapsed-day index + calendar one-hots

bar to beat — must beat it in backtests

simplest → most structure

requires_features = True
feature-aware · fit(y, X) · 5 models

bagging · parallel
independent trees, averaged



random_forest
RandomForestRegressor · sklearn
flag · off

boosting · sequential
each tree fits the previous trees' residuals

regression — HistGradientBoostingRegressor — always on

lightgbm — lightgbm.LGBMRegressor — ml-lightgbm · flag · off

xgboost — xgboost.XGBRegressor — ml-xgboost · flag · off

linear additive · prophet_like · ProphetLikeForecaster

X (lag cells NaN) → SimpleImputer(median) → Ridge(cholesky)

decompose(X): ŷ = intercept + trend + seasonality + holiday_regressor

always on — pure scikit-learn — not the Meta Prophet package · no flag

Rule: baselines first — a model that can't beat seasonal_naive in backtests doesn't justify its cost.

review's "gradient-boosted regression" = model_type regression · flags: random_forest, lightgbm, xgboost · extras: ml-lightgbm, ml-xgboost